CMPE 2965 - A02 - GROUP 8
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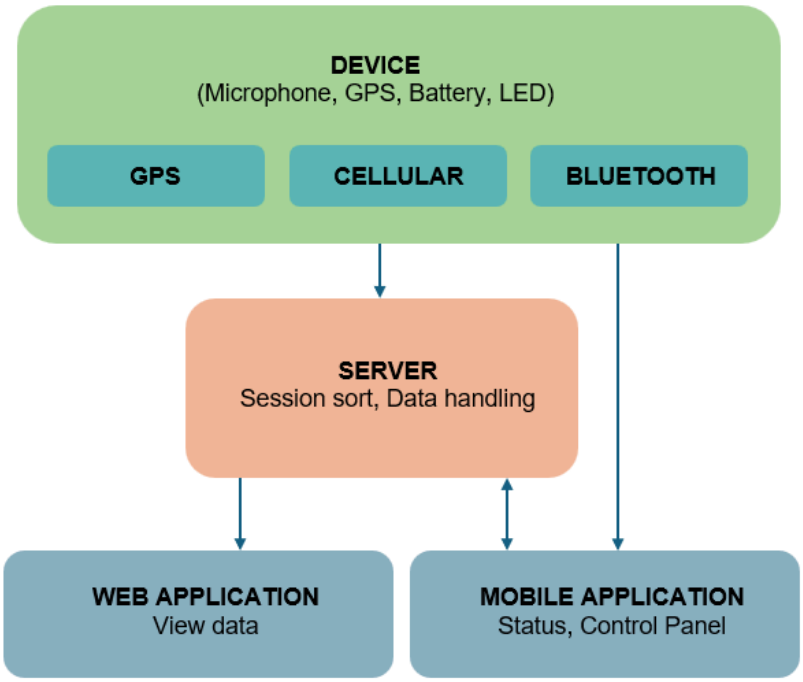
Project Purpose

Noise pollution is a concern for many organizations, and understanding where and when high levels occur is an important first step in managing it. This project aims to produce a portable monitoring device that collects sound level, location, and time data in industrial locations. The collected data will help identify areas of excessive exposure and assess patterns over time.

Proposed Solution

The device will sample ambient noise in decibels according to the standards set in ISO 1996-2:2017. Each data point will be tagged with date, time, GPS location, and the device ID number, and sent over a cellular connection to a server. The server will sort data into recording sessions based on date, time, and location. A mobile app will connect to the device via bluetooth to provide a device control panel to the user.

The device will have a power switch and a recording switch, along with a connection status button that activates LEDs indicating cellular, Bluetooth, and GPS position quality. The battery will have enough power for a couple days of operations and will be charged over USB-C. Battery level can be checked by pressing the battery button to activate indicator LEDs. The user will be able to select recording sessions to view and download through the web app.



Specification

Data Collection and Transmission

1. The device will sample ambient noise in dB once per second.
2. Measurements will be sent to the server over cellular connection, tagged with date, time, and location.
3. Data will be sorted into recording sessions by the server.

User Controls and Session Management

Users will be able to:

1. Turn the device on and off with a switch.
2. Start and stop data collection on the device using a switch on the device.
3. Select which recording session to view in the web application.
4. Manage the device through a mobile app control panel
5. Push the battery button to have the battery LEDs indicate current charge level
6. Push the connection status button to have the connection LEDs activate

Device Indicators

1. Connection quality will be indicated with three LEDs for cellular, bluetooth, and GPS.
2. Battery level will be indicated using LEDs at four charge thresholds: 25%, 50%, 75%, and 100%.

Connectivity

1. The device will use Bluetooth connectivity to connect to the control panel app.
2. The device will send data to the server over the cellular network.
3. The device will receive data for GNSS satellites for time stamping and location data

Mobile Application

The mobile app will:

1. Display status information about the device and the current recording session.
2. Send date, time, and location data to the device when GPS connection is unavailable (ex. inside a building)

Power

1. The device will be powered by a LiPo battery.

Innovation Strategy

Building this device is expected to require working with several technical areas that fall outside the standard CNT curriculum, requiring self-directed learning and applied research beyond standard coursework. Key areas include:

- **Cellular IoT communication**
- **GPS integration**
- **Mobile app development**
- **Bluetooth connectivity**
- **Low power consumption embedded development**

Knowledge and Skills Gap

The technologies listed in the Innovation Strategy section are all ones we do not have experience working with in the CNT program and we will need to learn how to use them, and we will need to familiarize ourselves with procedures for handling, processing, and interpreting audio data, as well as potentially improving our skills with analog electronics if we are unable to find a suitable existing sound sampling module.

As this device is meant to be deployed and left alone unmonitored, the embedded software will need to be written to be robust and stable, including error handling, which is not something we have been taught in the CNT program.

Resource Requirements

Hardware

- Microcontroller with cellular modem support
- GPS module and antenna
- Microphone/sound sensor and associated analog processing electronics
- Bluetooth Low Energy module and antenna
- LiPo battery and charging circuit
- Status LEDs, power switch, recording switch, battery and connection status buttons
- Enclosure materials (PLA 3D printer filament)
- Basic prototyping supplies (breadboards, wiring, soldering equipment, misc. electronic components)

Connectivity/Services

- IoT SIM card and cellular data plan (low-data tier sufficient for testing)
- Server hosting (cloud instance or on-campus server access) for the backend
- Domain/hosting for the web app

Software/Development Tools

- Firmware development environment
- Backend development framework and database
- Mobile app development framework
- Version control (Git/GitHub) for collaborative development

Budget Estimation

- Estimated total: $150 for one functional prototype device.

Timeline:

WEEK	ACTIVITY
1	● Course Introduction ● Project Brainstorming Exercise
2	● Project proposal draft and final
3	● Project plan and proposal presentation ● Technical specifications and verification (Draft)
4	● Technical specifications and verification (Final) ● SDLC assignment and Ethics assignment
5	● Hardware design draft ● Order parts for prototyping
6	● Finalize hardware design ● Begin software design
7	● Build app and server code ● Establish app-server communication
8	● Build and test prototype on breadboard, start embedded software
9	● System integration (device, server, and app) ● Project Synopsis and Progress report due
10	● Continue system integration
11 (Fall Break)	● Refine mobile app ● Begin web dashboard development
12	● Testing and polish ● Produce enclosure (3D-printed, if time permits) ● Work on user manual ● Final technical report (draft)
13	● Prepare final report and presentation ● Integration testing deadline ● User manual due
14	● Finalize presentation materials ● Final Technical Report submission
15	● Final presentation rehearsals ● Final presentation ● Project demonstrations